

Module 3: Strategic Equilibrium: Foundations & Applications

ECO364-THINKING STRATEGICALLY

Part II: Nash Equilibrium

DEFINITION & EXAMPLES

Nash Equilibrium

- ▶ All the previous examples could be solved by elimination of dominated strategies
- ▶ But there are situations where none of the strategies are dominated or none of the remaining strategies (after successive elimination of dominated strategies) are dominated and the game still has more than one outcome
- ▶ The next thing to do is to use the well known concept of strategic equilibrium (Nash 1950) which is known as Nash equilibrium



Nash, when applying to graduate school, in 1948



Nash, in the International Game Theory Conference at Stony Brook, 2014

Nash Equilibrium: Definition

- ▶ A **Nash equilibrium** is a collection of strategies (profile of strategies), one strategy for every player, such that no player has incentive to unilaterally deviate from his strategy (given that all other players stick to their strategies) .
- ▶ If a strategy s_i for player i belongs to a Nash Equilibrium profile, then this strategy is a best response to the strategies of the other players specified by the profile.
- ▶ In this sense, a Nash equilibrium is a “correspondence of best responses”.
- ▶ If you “conjecture” that your opponent is going to play a strategy that is part of a Nash equilibrium, then you should also be playing the corresponding strategy of said Nash equilibrium profile.

Nash Equilibrium: foundations

Common knowledge of rationality/conjectures

- ▶ If you read carefully the last slide, I mentioned something about conjectures: “if you conjecture (believe) that your opponent will play a strategy that belongs to a NE profile, then you should play the strategy that is prescribed to you in this profile”.
- ▶ As a solution concept, NE is stronger than iterated deletion of dominated strategies.
- ▶ Recall that iterated deletion of dominated strategies requires assuming Rationality of the players + Common Knowledge of Rationality.
- ▶ As in the last example of the Part I, sometimes this doesn't yield a unique strategy profile.
- ▶ Nash Equilibrium requires a little bit more: Rationality + Common Knowledge of Rationality + Common Knowledge of conjectures
- ▶ This means that the conjecture of which profile will be played must be common knowledge.
- ▶ This topic is a little advanced, so this is as far as we will pursue it here.
- ▶ Just bear in mind that the NE concept is stronger precisely because it requires Common Knowledge of Conjectures.

Example 1: 3x3 two-person game

		2		
		B ₁	B ₂	B ₃
1	A ₁	2, 0	3, 3	4, 2
	A ₂	5, 5★	4, 4	1, 4
	A ₃	3, 5	6, 6	2, 7

- ▶ Let's consider the situation first from the perspective of player 1.
 - ▶ If player 2 plays B₁, then player 1 should respond by playing A₂ because in that way, she maximizes her payoff. **A₂ is a best response to B₁.**
 - ▶ If player 2 plays B₂, then player 1 should respond by playing A₃ because in that way, she maximizes her payoff. **A₃ is a best response to B₂.**
 - ▶ If player 2 plays B₃ then player 1 should respond by playing A₁ because in that way, she maximizes her payoff. **A₁ is a best response to B₃.**
- ▶ Let's consider the situation first from the perspective of player 2
 - ▶ If player 1 plays A₁ then B₂ is a best response.
 - ▶ If player 1 plays A₂ then B₁ is a best response.
 - ▶ If player 1 plays A₃ then B₃ is a best response.
- ▶ Then, the profile (A₂, B₁) is a Nash Equilibrium. It is a correspondence of Best responses. If 1 conjectures that 2 will play B₁ then 1 should play the strategy specified by the profile, A₂ because it is a best response.
- ▶ Similarly for player 2, if she conjectures that player 1 will play A₂, then 2 should play the strategy specified by the profile, B₁ because it is a best response.
- ▶ At the NE, the conjectures of what the opponent play turn out to be correct. Player 1 correctly conjectures that player 2 will play B₁ and player 2 correctly conjectures that player 1 will play A₂.
- ▶ At the NE equilibrium point, no deviation from it is profitable.
- ▶ Note that (5, 5) is inferior to the outcome (6, 6) which is obtained when 1 chooses A₃ and 2 chooses B₂
- ▶ At the profile (A₃, B₂). In here, player 2 has an optimal deviation, which is to play B₃ and get 7 instead of 6. Thus this profile is not a Nash equilibrium.

Example 2: 2x2 two-person game

		2	
		L	R
1	U	2, 2★	0, 0
	D	0, 0	1, 1★

- Here there are two Nash equilibrium points
(U, L) with the outcome (2, 2) and (D, R) with the outcome (1, 1)

Example 3: Stag hunt

- ▶ Each of two hunters can either try to hunt a stag (an adult deer and rather large meal) or hunt a rabbit (tasty, but substantially less filling). If an individual hunts a stag, he must have the cooperation of his partner in order to succeed. However, a hunter can easily get a rabbit by himself. Each player must choose an action.
- ▶ The NE are the profiles (stag, stag) and (rabbit, rabbit).
- ▶ Each hunter must coordinate with the other if they are going to be successful.
- ▶ In this strategic game, the players are not at conflict with each other. **They need to cooperate.**
- ▶ A strategic game can also describe this type of situation.
- ▶ Going back to the Grade Game from last week, of both you and your partner had preferences for fairness, you would achieve to a game matrix similar to this one. You would need to coordinate to play the same strategy.

		Hunter II	
		stag	rabbit
Hunter I	stag	4,4	0,2
	rabbit	2,0	2,2

Example 4: Game of chicken

- ▶ The name of this game comes from the movie “Rebel without a cause” from 1955. [Clip of the movie](#)
- ▶ This is a simplified version. Two players (read, teenage boys) decide to settle a dispute (read, teenage girl) by way of near-death experiences. Each speeds an automobile towards a cliff (or at each other, in later adaptations). A simple rule governs the challenge: the first to jump out of his automobile is the chicken and, by universally accepted social convention, concedes the object in dispute. The second to jump (tough) is victorious, and, depending on context, becomes gang leader, prom king, etc.
- ▶ Of course, if both jump simultaneously, a stalemate results, and if neither jumps, a fiery death is assured since all cars (in movies) explode upon impact. Whether the dead are celebrated by their peers is inconsequential, since game theorists are apt to assume that the payoffs from this outcome are quite negative.
- ▶ The game as described has two equilibria (tough, chicken) and (chicken, tough).
- ▶ The game is also called dove-hawk in biology.
- ▶ You also face this game when entering a junction!

		player II	
		tough	chicken
player I	tough	-10,-10	6,0
	chicken	0,6	3,3

Example 5: 5x5 two-person game

- ▶ There are exactly two Nash equilibrium points:
- ▶ (A_1, B_4) with the outcome $(8, 10)$
- ▶ and (A_3, B_3) with the outcome $(10, 9)$
- ▶ Is there an easy way to find all Nash equilibrium points of a game?
The answer is YES
- ▶ Here is the procedure:
 - ▶ For every strategy of player 1 (every row) designate by * the upper right side of the square containing the highest payoff of player 2
 - ▶ For the row A1 we designated the two squares $(A1, B2)$ and $(A1, B4)$ where 2 obtains 10
 - ▶ For the row A2 we designated the square $(A2, B3)$ where 2 obtains 12, etc.
 - ▶ Next, for every strategy of player 2 (every column) we designated by * the upper left side of the square containing the highest payoff of 1
 - ▶ For the column B1 we designated the square $(A4, B1)$ where 1 obtains 14
 - ▶ For the column B2 we designated the square $(A2, B2)$ where 1 obtains 15, etc.
 - ▶ The squares that are designated twice represent the Nash equilibrium points
 - ▶ In the example above these squares are $(A1, B4)$ and $(A3, B3)$

		2				
		1				
		B ₁	B ₂	B ₃	B ₄	B ₅
A ₁		10, 5	5, 10 *	7, 7	* 8, 10 *	6, -3
A ₂		4, 9	* 15, 3	4, 12	6, 10 *	9, 9
A ₃		8, 7	7, 8	* 10, 9 *	7, 8	* 15, 7
A ₄		* 14, 11	12, 13	* 10, 9	5, 5	5, 14 *
A ₅		6, 9 *	9, 6	8, 8	7, 7	10, 6

Example 6a: Eliminating weakly dominated strategies can delete an equilibrium

- ▶ The strategies b and β are **weakly dominated**.
- ▶ After eliminating these strategies we are left with the strategy combination (a, α) , which is an equilibrium of the original game.
- ▶ The combination (b, β) is an equilibrium of the original game that was eliminated.
- ▶ A Nash Equilibrium thus might consist of weakly dominated strategies.
- ▶ Be careful here! We are talking about weakly dominated strategies, not strictly dominated ones!

	α	β
a	★ 1,1 ★	0,0
b	0,0	★ 0,0 ★

Example 6b: Eliminating weakly dominated strategies can delete an equilibrium

- Consider the following 3x3 two-person game:

	B ₁	B ₂	B ₃
A ₁	★ 1, 1 ★	0, 1	0, 1
A ₂	1, 0	1, 0	0, 1
A ₃	1, 0	0, 1	1, 0

- The only Nash equilibrium here is (A₁, B₁) with the outcome (1, 1)
- This outcome is the best outcome in the table **but it is not sensible**
- Note that A₁ is (weakly) dominated by both A₂ and A₃ and B₁ is (weakly) dominated by both B₂ and B₃
- We see that a Nash equilibrium may consist of weakly dominated strategies!

Example 6b: Eliminating weakly dominated strategies can delete an equilibrium

- Consider the following 3x3 two-person game:

		2		
		B ₁	B ₂	B ₃
1	A ₁	★ 1, 1 ★	0, 1	0, 1
	A ₂	1, 0	1, 0	0, 1
	A ₃	1, 0	0, 1	1, 0

- ▶ No strictly dominated strategy can be part of a Nash equilibrium (why?)
- ▶ Recall that in the previous part I mentioned that a **strategy is a best response if and only if it is not strictly dominated by another strategy**.
- ▶ This means that a strategy that is strictly dominated is never a best response and vice versa.
- ▶ For this reason, no strictly dominated strategy can be part of a NE, since a NE is a correspondence of best responses.
- ▶ However, from this example notice the strong requirement that there should be common knowledge of conjectures. Player 1 should believe that 2 will play B₁ and player 2 should believe that 1 will play A₁ for (A₁, B₁) to arise as an equilibrium. But why should it be, since the strategies that are involved in this equilibrium are weakly dominated!
- ▶ Let us also mention that this game has other Nash equilibrium points, but in mixed strategies (strategies that are selected by a random device) which do not use any weakly dominated strategies
- ▶ Every player uses the strategies (0, 1/2, 1/2) which means that he chooses at random one of his last two strategies, equally likely
- ▶ We will study mixed strategies later on